# 1. Clinical Question & Core Hypothesis

## Clinical Question

Immunologically 'hot' NMIBC (high TIL density) does not uniformly benefit from immune checkpoint therapies — why does immune infiltration fail to translate into clinical response in a subset of tumors? 2) Tier 1 paths used and why: I built on the MSI/dMMR path because mismatch-repair defects directly raise neoantigen burden and are mechanistically linked to immunotherapy responsiveness; I also used the TMB path because mutation load quantifies neoantigenic potential and can modulate the functional capacity of infiltrating T cells. I deprioritized Immunoscore as a primary anchor to keep the hypothesis sharply molecular (neoantigen supply) while using an H&E-derived immune metric (TIL density) as the imaging readout. 3) Tier 2 evidence that supports the claim: eIF4E and GARS1 provide mechanistic modifiers of tumor immune phenotype (translation programs and noncanonical immune signaling that can shape antigen presentation or immune suppression); urine DNA methylation biomarkers and CTCs offer noninvasive surrogates for tumor state and could identify discordant cases (TIL-high but low neoantigen load); HDL-C and MSH6 add metabolic and gene-specific nuance to immune–genomic interactions. These supporting paths justify testing whether neoantigen supply (MSI/TMB) conditions the effectiveness of local immune infiltrates. 4) Why I chose therapy_response as the endpoint: The core tier-1 paths repeatedly converge on biomarkers (MSI/dMMR, TMB, Immunoscore) that are most relevant to predicting benefit from immunotherapy, making therapy response the most mechanistically pertinent and clinically actionable endpoint for NMIBC. 5) What I deliberately EXCLUDED and why: I excluded using Immunoscore as a primary anchor to obey the Focusing Rule (max 2 anchors) and because TIL density on H&E is a simpler, directly measurable imaging surrogate in many clinical cohorts; I excluded wide panels of additional genes (e.g., SARIFA, broader metabolic panels) from the hypothesis sentence to avoid a model-driven, unfocused claim — these sit in secondary_modifiers for follow-up analyses?

## Working Hypothesis

In NMIBC, tumor-infiltrating lymphocyte density does not predict benefit from immune checkpoint blockade when tumors lack mismatch-repair deficiency and have low tumor mutational burden, because neoantigen load is insufficient to sustain effective T-cell–mediated tumor clearance. Therefore, NMIBC patients with high TIL density and high neoantigen burden (as indicated by MSI/dMMR or elevated TMB) will have improved responses to immunotherapy, whereas those with high TIL density but low neoantigen burden will exhibit poor responses.

| Element | Value |
|---|---|
| Imaging feature | Tumor-Infiltrating Lymphocyte (TIL) density |
| Anchor gene(s) | MSI/dMMR, TMB |
| Primary endpoint | therapy_response |

# 2. Why This Gene — Not Others

## Primary hypothesis gene(s)

**MSI/dMMR:** MSI/dMMR path because mismatch-repair defects directly raise neoantigen burden and are mechanistically linked to immunotherapy responsiveness; I also used the TMB path because mutation load quantifies neoantigenic potential and can modulate the functional capacity of infiltrating T cells.

**TMB:** TMB, Immunoscore) that are most relevant to predicting benefit from immunotherapy, making therapy response the most mechanistically pertinent and clinically actionable endpoint for NMIBC.

## Alternative / secondary genes considered

| Tier | Gene | Role / Reason excluded from core |
|---|---|---|
| Secondary | Immunoscore | Supporting modifier from knowledge graph |
| Secondary | eIF4E | Supporting modifier from knowledge graph |
| Secondary | GARS1 | Supporting modifier from knowledge graph |
| Secondary | urine DNA methylation biomarkers | Supporting modifier from knowledge graph |
| Secondary | CTCs (liquid biopsy) | Supporting modifier from knowledge graph |
| Secondary | HDL-C | Supporting modifier from knowledge graph |
| Secondary | MSH6 | Supporting modifier from knowledge graph |

## 3. Clinically Testable Predictions

**Prediction 1:** such as eIF4E and GARS1, urine DNA methylation markers, CTCs, HDL-C, and MSH6 as exploratory strata to refine patient selection, while maintaining the primary hypothesis anchored on MSI/dMMR and TMB. Limitations to address: heterogeneity of NMIBC and prior treatments; need for sufficiently powered cohorts; potential confounding immune microenvironment factors.

**Prediction 2:** In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

**Prediction 3:** At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

## 4. Minimum Viable Validation Plan

### A. Retrospective Cohort Analysis

**Data source:** TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

### B. Tissue-Level Validation

**Methods:** Multiplex IF / IHC / spatial transcriptomics

**Key readouts:** - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

**Goal:** Confirm that the proposed mechanism operates at the spatial / tissue level.

### C. Functional Validation

**Models:** Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

**Perturbations:** Genetic (KO/rescue) or pharmacologic modulation of anchor gene

**Readouts:** T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

**Goal:** Establish causal relationship between anchor gene activity and immune efficacy.

## 5. Strengths, Caveats & Alternative Directions

### Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

### Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

### Alternative Directions

- Immunoscore: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- eIF4E: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.